

Driven by Learning: Exploring Older Adults' Pedagogical Preferences in Highly Automated Vehicle Interactions

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Abstract

The learning phase for in-vehicle technologies is critical in the adoption of automated vehicles (AVs), particularly for older adults, who often rely on peer support and mentorship to navigate new technologies. We conducted a study involving nine older adults who experienced a Level 4 highly automated vehicle on a closed circuit track, followed by a workshop. Using a participatory design approach, the participants visualized their ideal human-machine interface (HMI) designs and expressed how external and mutual support among older adults can facilitate learning. Our preliminary findings revealed a strong preference for a combination of experiential learning, structured guidance, clear instructional materials, and social learning. Through the mentorable interaction framework, we explore how AV systems can integrate mentorship and peer-assisted guidance. Our study underscores the need for AV systems that align with users' pedagogical needs.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Accessibility theory, concepts and paradigms**; • **Social and professional topics** → **Seniors**.

Keywords

older adults, automated vehicles, technology education, digital literacy, learning methods

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1 Introduction

The stereotype that older adults struggle with technology is both pervasive and deeply rooted in societal attitudes [20]. As automation and smart technologies become increasingly integrated into

daily life, older adults are often dismissed as unable to adapt to these innovations [25]. Over time, such stereotypes can become internalized, leading older adults to doubt their abilities and miss out on the potential benefits these technologies may offer [18, 24, 26].

As automation becomes more prevalent, particularly in tasks meant to enhance convenience and accessibility, it is vital to consider how older adults perceive and engage with these advancements. Automated vehicles (AVs) represent a notable example of this innovation, with significant implications for improving mobility and independence [11]. Although AVs are often touted as transformative tools in this regard, there is still a lack of clear understanding of how users will interact with AVs in practice.

The initial learning phase of technology adoption can be particularly challenging for older adults, who rely on external and peer support to navigate new systems. It is especially relevant since older adults have been found to value learning and support most in everyday technologies [10]. In a previous study, we introduced the concept of “mentorability,” where mentorship is integrated into the learnability of technologies and considerations are given to instrumental and psychosocial support [5]. Building on this framework, we analyze the perspectives of older adults through participatory design workshops following experiences in a Level 4 automated vehicle. Based on their first-time experiences in AV usage, participants designed their preferred interfaces in the workshop and brainstormed interaction scenarios for AV systems, expressing diverse concerns about the use of AVs and their impact on their daily lives. Their responses emphasized the critical need for instructional and psychosocial support in fostering effective engagement with AVs.

The reliance on peer and external support among older adults raises important questions about how such peer support might take shape in the context of AV adoption. This late-breaking work presents preliminary findings on the unique pedagogical needs of older adults and lays the groundwork for ensuring that future AV designs are not only accessible but also empowering for this demographic.

2 Background

Older adults face unique challenges and opportunities when interacting with AVs. These technologies offer new opportunities for social engagement by enabling older adults to maintain mobility and independence, which are crucial to their well-being [11, 19, 21]. AVs address challenges associated with age-related declines in cognitive and physical functions, allowing them to continue participating in social, community, and family activities [12]. By reducing reliance

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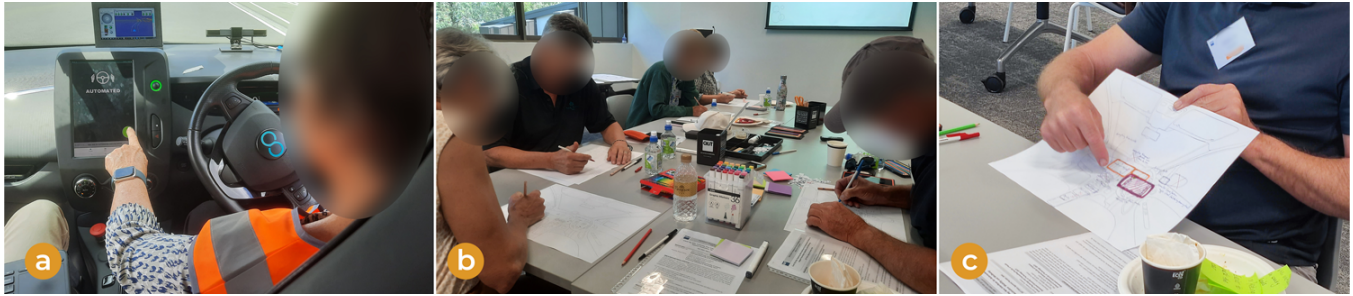


Figure 1: (a) Participant in the L4 automated vehicle engaging automated mode (b) participants during the workshop and (c) a participant demonstrating their preferred HMI design

on traditional driving skills, AVs can not only mitigate safety risks [23] but also improve quality of life, which has often been linked to driving cessation [8].

However, barriers to AV adoption among older adults remain significant. Concerns about unfamiliarity with technology, safety, reliability, and the absence of tailored user interfaces are prominent challenges [1]. Previous research has highlighted the importance of trust, situational awareness, and thoughtfully designed human-machine interfaces (HMIs) in fostering users' comfort and confidence in AV systems [3, 28]. To address these concerns, including older adults in the design process may ensure AV technologies align with their specific needs and capabilities [2, 5].

A critical factor in the adoption of AVs is the ability of older adults to learn and interact effectively with in-vehicle technologies [27]. Unlike younger users, older adults often favor structured guidance, peer support, and experiential learning methods over trial-and-error approaches [5, 15]. These preferences are shaped by broader cognitive and emotional dynamics, such as the need for confidence-building and clear communication of complex functionalities [12]. Social learning, mentorship, and high-quality instruction significantly enhance their ability to engage with new technologies and eventual adoption [4, 7]. To leverage the associated peer support and mentorship, a framework for "mentorability" was previously proposed that takes into account the instrumental and psychosocial support that is often overlooked in the main principles of learnability [5]. By addressing diverse learning styles, promoting human-centered pedagogical designs through such a framework, and fostering trust [13], AV systems may empower older adults to engage confidently with AVs.

3 Procedure

To explore the support necessary for older adults in the use of AVs, we conducted a study that consisted of two main components: (1) a driving trial, as shown in Figure 1 (a), where we invited each older adult participant to experience an AV in the driving seat, with a safety driver in the passenger seat, and (2) a post-drive workshop, as shown in Figure 1 (b, c), where four to five participants reflected on their experiences as a group, discussed their concerns and challenges with automation, and collaboratively designed and proposed potential interventions through an ideation process.

Our research questions aimed to examine how older adults interact with highly automated vehicles (HAVs) and what forms of

support they require for effective learning and adoption. Specifically, we investigated the following:

- How do older adults prefer to learn about and interact with HAVs?
- What role does peer support and mentorship play in facilitating older adults' understanding and adoption of technology automation and HAVs?
- How can human-machine interfaces (HMIs) be designed to align with the pedagogical needs of older adults?

3.1 Participants

We recruited nine older adult participants through the distribution of flyers at a local senior activity club and through snowball sampling, where initial participants referred others from their social circles, including members of a local driving club. Participants were required to be 50 years of age or older while identifying as 'older adults' to be eligible. This criterion was chosen due to the inconsistency in age thresholds used by various agencies to define the demographic [5]. Because the study involved a driving component, participants were required to hold valid driver's licenses following the equipment operation protocol.

The participants (5 male, 4 female) ranged in age from 56 to 76 ($M=66$; $SD=5.88$). We collected participant information relating to their background in driving, including how often they drive or take public transportation, their residential regions, and their years of driving experience. For their role in the study, all participants were given gift cards for a local supermarket chain store upon completing the driving trial and workshop components. The list of participants can be seen in Table 1.

3.2 Equipment

For the driving component, we used a modified Renault ZOE that is an experimental highly automated vehicle (HAV), capable of level 4 automation as designated by the Society of Automotive Engineers [17], and is capable of traveling up to 50km/h in automated driving mode on designated public roads as per current local laws and regulations. In our study, the AV operated on a closed circuit track. As seen in Figure 2, the HMI displayed a single screen showing the current mode of operation (automated or manual).

Table 1: Participants' demographics

Participant ID	Age	Gender	Years Driving	Residential region	Frequency in driving	Workshop
P1	66	Male	50	Suburban	Very frequent	W1
P2	76	Female	55	Suburban	Very frequent	W1
P3	73	Female	57	Urban	Frequent	W1
P4	64	Male	46	Suburban	Very frequent	W1
P5	61	Female	43	Suburban	Frequent	W1
P6	69	Male	52	Rural	Very frequent	W2
P7	61	Female	44	Urban	Frequent	W2
P8	66	Male	48	Suburban	Very frequent	W2
P9	56	Male	50+	Suburban	Very frequent	W2

3.3 Ethical considerations

We obtained ethical approval (approval number 8664) from the university's ethics committee and conducted a risk assessment. At the start of the study, participants underwent a safety induction, which included viewing a three-minute training video specifically tailored to align with the planned route and scenario. Participants were informed of their right to withdraw from the study at any point, including mid-drive, particularly if they experienced motion sickness. To ensure privacy and confidentiality, participants were assured that all collected data would be accessible only to the research team. Additionally, any identifying information, such as faces in photos and videos, would be blurred, and names would be anonymized. Written consent was obtained from all participants prior to their involvement in the study. Due to ethics and safety protocols, a safety driver was required to be in the passenger seat at all times while participants were in the driver seat.

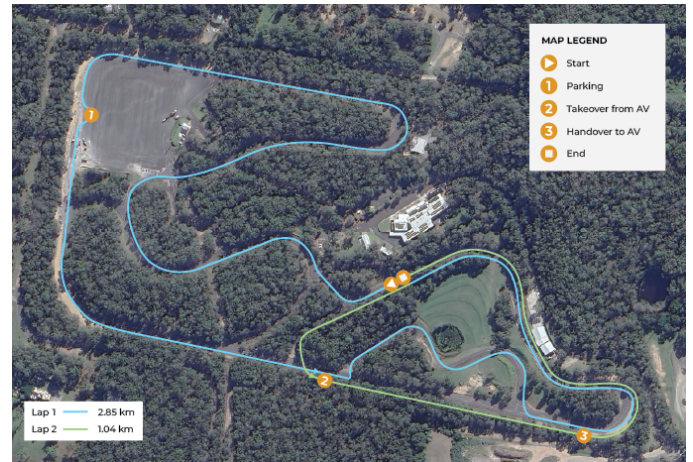
**Figure 2: Interior of the level 4 HAV**

3.4 Driving trial

Each participant completed a ten-minute session around the closed-circuit track, following a route designed in collaboration with the broader AV research team, including engineers in charge of the

vehicle, psychologists, human-factors and HMI experts. The route was designed to showcase the capabilities of a Level 4 HAV and included a take-over scenario, hand-over scenario, and a “parking” task, as shown in Figure 3.

The route consisted of two laps. The first lap involved the parking task, where participants had to engage the automated function and resume the pre-planned route by pressing a button on the HMI. The second, smaller lap included a manual driving portion, which lasted approximately 3 minutes.

**Figure 3: Automated vehicle route at the closed-circuit test track**

3.5 Workshop

Immediately after the driving trial, participants attended a three-hour workshop at the same location, which included reflection, ideation, and discussion. Each workshop included the following:

Introduction (10 min): The session began with a reiteration of the study's research goals, followed by an exploration of participants' immediate first-hand reactions to their AV experience.

Discussion 1 (40 min): Participants discussed vehicle automation, AV use cases, future transportation implications, and their

learning needs, contextualized by their daily transportation habits and technology usage.

Discussion 2 (45 min): To familiarize participants with AV concepts, videos of Waymo robo-taxis in San Francisco were shown, along with examples of emerging technologies like windshield displays, virtual assistants, and augmented reality avatars. These examples, informed by the "mentorability" framework, encouraged participants to envision enhanced AV user experiences.

Ideation and Discussion 3 (80 min): Participants engaged in a hands-on session, where they drew over images of the ZOE interior and designed their preferred HMIs for AVs. They presented their concepts to the group, highlighting functionality needs and benefits.

3.6 Data collection

Both the driving trials and workshop sessions were audio and video recorded, and the recordings were transcribed using transcription software. For qualitative analysis, we employed a reflexive thematic analysis [6]. The transcriptions were systematically reviewed, and themes were identified and coded using a Miro board. This approach facilitated the organization and visualization of emerging patterns, enabling a deeper understanding of participants' experiences and insights.

4 Findings

This section summarizes the preliminary findings from participant responses during the workshop sessions. It highlights their experiences, attitudes, and perceptions of AVs, shedding light on both the challenges faced and the opportunities identified. Each subsection explores key themes that emerged, offering an interim understanding of user needs, preferences, and concerns. These insights provide a foundational perspective to inform ongoing discussions on AV design, user education, and policy considerations.

4.1 User Education and Learning Preferences

Participants expressed a strong preference for hands-on learning methods over theoretical approaches. Familiarity with AV features, such as buttons and controls, was seen by participants as essential to reducing apprehension. Video-based tutorials and guided trials were suggested as effective ways to facilitate this learning process.

Regarding learning materials, participants stated that they should avoid using technical terminologies and rather informal means prior to using the AV. As P4 noted, learning materials should be:

Very simple, very visual. I think you've got to not use acronyms like AI if you don't know what AI stands for, what does it mean? So you've got to take, a lot of computer people or tech savvy people use all their acronyms, or government people. You've got to bring it back to the basics because you've never done this before. So I think it's really important.

When asked which sources participants would receive information from, they indicated they would rely on family members or peers to assist in learning AV technology. This was exemplified by one participant who recalled asking their son for help with a ride-hailing app, suggesting similar mentorship models could be valuable

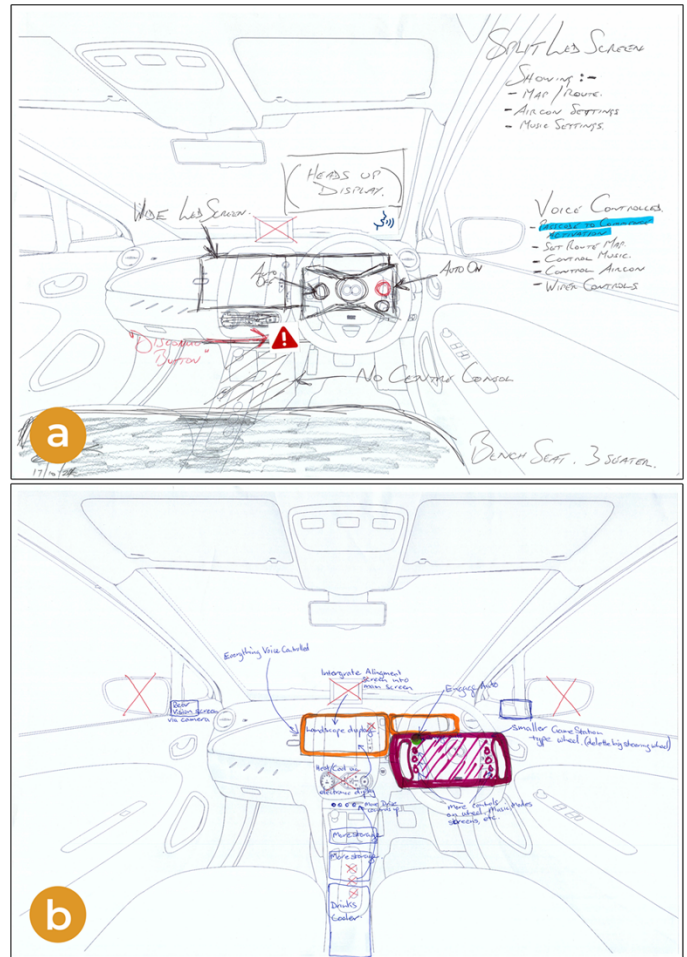


Figure 4: Sketches by P1 (a) and P4 (b)

for AVs. Generational differences were evident, with younger individuals perceived as more adaptable due to their greater exposure to technology, however the participants expressed that younger people were "impatient" with older learners. The participants acknowledged a steeper learning curve influenced by established driving habits and largely preferred an induction session embedded into the system or HMI, that demonstrates the functions and operations.

4.2 Interface Design and Interaction Preferences

Participants provided several design suggestions to improve the HMIs of AVs. Key controls, such as activation buttons were recommended to be placed on the steering wheel or within easy reach to minimize distraction as seen in Figure 4 (a).

When we provided the participants with five interaction modality examples (gaze, touch, gesture, voice, hearing), consistent with previous research [16], voice control emerged as the preferred modality for interacting with AVs, with participants highlighting its

convenience and suitability for scenarios requiring route alterations, as shown in Figure 4 (b).

Additionally, similar to previous studies [16, 22], participants showed keen observations and concern regarding cyber security, stating that they would like to have a 'kill switch' or a 'just-in-case button' that would allow the user to take over the vehicle. As P1 noted:

If these things are fully automated and computerised there, how safe and secure is that is going to be? Even, especially in the future where hacking is becoming more common and you want to be more reliant on those types of systems as well. That's why I want a kill switch on.

The majority of participants expressed interest in emerging technologies like AI companions and virtual assistants that could learn and adapt to user preferences. For example, an AV that could teach and walk users through the functions while predicting alternative routes based on traffic conditions was seen as desirable.

I think a virtual assistant's probably a good thing. And that could be on a computer screen as well. And you hop into the car and someone will say to you, so where, where do you want to go to today?

4.3 Attitudes and Perceptions Toward AVs

First-hand reactions upon completion of the driving trial were positive overall. However, as the AV was an experimental prototype rather than a commercially available model, participants noted its "clunky" and "mechanical" nature. The vehicle's movements were often abrupt, creating a sense of unpredictability during the ride. Unlike commercially available AVs, the research vehicle did not provide a view of future vehicle trajectory. As a result, several participants were surprised when the vehicle unexpectedly chose a direction, catching them off guard and raising concerns about the system's decision-making processes.

Participants described their experience with the AV as feeling more like being a 'passenger' rather than a 'driver' and compared it to driving in another country, where one sits on the opposite side of the car, emphasizing the unfamiliarity of the experience. Initially, many participants expressed apprehension about entering and using the AV, unsure of what to expect. However, as the ride progressed, they became more aware of the vehicle's capabilities and limitations, which gradually fostered a sense of trust in the technology.

A significant challenge highlighted by participants was the transition between manual and automated modes. The participants occasionally struggled with hand-over processes, leading to moments of discomfort, and underscored the importance of a reliable and intuitive transition process that feels safe and more controllable.

Despite these challenges, participants expressed a willingness to adapt to AVs after repeated use, noting that exposure could alleviate initial apprehensions. However, some voiced concerns about the potential loss of the pleasure of driving, emphasizing how central driving is to their identity and daily life. The ability to drive was described as a crucial part of self-esteem and independence.

The discussion revealed that consistent and positive exposure to AV technology could help mitigate initial apprehension, particularly in rural settings where the technology may feel less familiar. Participants acknowledged the potential of AVs to enhance accessibility and independence, even as they navigated concerns about the inevitable loss of driving ability and its impact on their lives. As P2 noted:

Having a driver's licence is part of your identity. Once you lose that, whether it's old age or medical conditions, but you've lost part of yourself, you really do feel you've aged. But if you had access to an AV, it would change everything, wouldn't it?

The unique challenges of implementing AVs in Australia emerged as a key concern in both workshops. Participants noted the vast rural areas, damaged roads, and inadequate infrastructure that could hinder widespread adoption. Discussions on rural and urban usage revealed a preference for AVs in urban environments. Rural areas were perceived as posing significant challenges due to unpredictable road conditions and the joy many participants derive from country driving. Concerns about other drivers' predicted behaviors also surfaced as participants voiced concerns regarding safety and interaction with other vehicles.

The participants envisioned AVs as useful for specific scenarios, such as shopping or monotonous driving, and stated that AVs could be particularly beneficial for individuals unable to drive due to disabilities. A suggestion was made in workshop 1 (W1), to introduce a specialized 'AV license' for individuals who may not qualify for conventional driving licenses but could safely use AVs.

In particular, the concern for policy implementation emerged as a key topic, as participants highlighted that significant restructuring was needed to accommodate large-scale AV implementation.

5 Discussion

Participants demonstrated cautious optimism toward AVs, recognizing their potential to enhance mobility and independence while identifying key challenges in design, infrastructure, and user experience. To address these issues, our findings emphasize the importance of learnable interfaces, structured guidance, and systems tailored to diverse contexts and preferences.

Hands-on experiential learning with clear and structured guidance was a consistent preference, reflecting the need of older adults for clarity and contextual support. As with the "mentorability" framework, understanding the psychosocial and instrumental support, including how and when to receive the learning materials, and how they influence the users' motivation to learn and use AVs was crucial to the potential adoption of AVs. As participants voiced a preference of using YouTube videos when learning new technologies, video tutorials and guided practice sessions were identified as effective strategies. Through peer-assisted learning programs, such as mentorship or community workshops, and fostering a shared learning environment [9], the adoption of and trust associated with AVs could potentially be increased. This is supported by previous studies, where it was found that information regarding AV technology can influence users' perceptions and attitudes toward AV acceptance [14].

Voice control emerged as the preferred interaction modality for its convenience. Participants, particularly those with higher educational attainment, expressed openness to using conversational AI agents as a learning tool for AV technology, provided they remain simple and intuitive. For future studies, AI assistants trained on user-generated data and providing video-based support could facilitate engagement and adoption, particularly if they are designed to align with older adults' personal usages. This is particularly of note as some participants favored private AVs for autonomy and personalization, while others preferred shared robo-taxis for their practicality. Considering this, future AV systems should accommodate both models and tailor HMIs accordingly, balancing the need for personalization in private ownership with accessibility and ease of use in shared systems.

Incorporating these insights into AV design can ensure that these technologies are not only accessible but also empowering for older adults, supporting their independence and mobility across diverse contexts.

6 Limitations

This study had several limitations that should be addressed in future research. First, a safety driver was required to be present in the vehicle with the participant for protocol and safety reasons. It would be valuable to investigate how participants might interact with the AV if they were alone, as this could provide a more authentic representation of real-world use.

Second, the study was conducted with nine participants in a major city in Australia, which may limit the generalizability of findings. Exploring regional differences, particularly between urban and rural settings, could offer additional insights into how automated vehicles are perceived and utilized.

Third, the vehicle used in this study is an experimental Level 4 highly automated prototype. As such, it exhibited some technical limitations and operated on a closed track on a pre-planned route without interactions with other vehicles. As commercial AVs are not available for use in many parts of the world, we conducted the study on a closed circuit track rather than an open environment with real-world traffic conditions.

Finally, participants in this study were required to hold a valid driver's license, which excluded individuals who no longer drive due to age, disability, or other reasons. Including non-drivers in future research could provide critical insights into the needs and preferences of this demographic, who stand to benefit significantly from AV technology.

7 Conclusion and Future work

The test drive offered participants an opportunity to gain a deeper understanding of AV development and its practical applications. This experience provided valuable insights into the potential benefits and challenges of the new technology. Following the test drive, the workshops facilitated an environment where participants could engage in the creative process of conceptualizing and designing potential HMIs based on their experience. These sessions encouraged participants to contribute their ideas and perspectives, which could be further refined into prototypes for future studies. This collaborative approach helped to explore how interfaces might be effectively

tailored to enhance user experience for older adults interacting with AV technology.

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